A Place to Learn, A Place to Grow

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# JIPMAT 2022

## QUESTION PAPER WITH SOLUTIONS

## QUANTITATIVE ABILITY

1. A sum becomes double in 10 years, then annual rate of simple interest is
   (a) 8%  (b) 5%
   (c) 10%  (d) 20%

2. P, Q and R invested Rs. 25,000, Rs. 50,000 and Rs. 25,000 respectively to start a business. At the end of two years, they earned a profit of Rs. 48,000. What will be Q's share?
   (a) 24,000  (b) 36,000
   (c) 12,000  (d) 20,000

3. Match List I with List II
   Choose the answer from the given below:

| List I(Quadratic Equation) | List II(Roots) |
|---|---|
| $A. 6x^2 + x - 12 = 0$ | $I. (-6, 4)$ |
| $B. 8x^2 + 16x - 10 = 202$ | $II. (9, 36)$ |
| $C. x^2 + 45x + 324 = 0$ | $III. \left(3, -\frac{1}{2}\right)$ |
| $D. 2x^2 - 5x - 3 = 0$ | $IV. \left(-\frac{3}{2}, \frac{4}{3}\right)$ |

   (a) A-I, B-III, C-IV, D-II  (b) A-IV, B-I, C-II, D-III
   (c) A-II, B-IV, C-III, D-I  (d) A-III, B-II, C-I, D-IV

4. The population of a village is 5000. If in a year, the number of males were to increase by 5% and that of females by 3% annually, the population would grow to 5202 at the end of the year. If M is the number of males and F is the number of females in the village, then (M, F) =
   (a) (3000, 2000)  (b) (2800, 2200)
   (c) (2600, 2200)  (d) (2700, 2300)

5. If A earns $33\frac{1}{3}$% more than B, then percentage B earns less than A will be
   (a) $66\frac{2}{3}$%  (b) 25%
   (c) 65%  (d) $33\frac{1}{3}$%

6. Which of the following trigonometric identities are true?
   A. $\sin^2(41°) + \sin^2(49°) = 1$
   B. $\sin^2(60°) - 2\tan(45°) - \cos^2(30°) = -1$
   C. $\sin^2(\theta) + \frac{1}{1+\tan^2(\theta)} = 1$
   Choose the correct answer from the options given below:
   (a) A and B only  (b) A and C only
   (c) B and C only  (d) A, B and C

7. Which of the following is the value of m for which the polynomial $x^4 + 10x^3 + 25x^2 + 15x + m$ is exactly divisible by $x + 7$ ?

(a) -91 (b) -101

(c) 15 (d) 115

8. The area of a circular play ground is 22176 $Cm^2$ . What is the cost of fencing this ground at the rate of ₹50 per metre?

(a) ₹ 264 (b) ₹ 400

(c) ₹312 (d) ₹ 244

9. The sum of n-terms of sequence $\frac{1}{1\times2}\times\frac{1}{2\times3}\times\frac{1}{3\times4}\times-$..........is

(a) $\frac{1}{n+1}$ (b) $\frac{1}{n}$

(c) $\frac{n+1}{n}$ (d) $\frac{n}{n+1}$

10. If A can do $\frac{1}{4}$ of a work in 3 days and ' B ' can do $\frac{1}{6}$ of same work in 4 days, how much will, 'A' get if both work together and are paid ₹180 in all

(a) ₹36 (b) ₹ 60

(c) ₹108 (d) ₹120

11. Two poles of height 6 m and 11 m stand vertically upright on a plane ground. If the distance between their foot is 12 m , then distance between their tops is

(a) 12 m (b) 14 m

(c) 13 m (d) 11 m

12. The value of a flat worth ₹5,00,000 is depreciating at the rate of 10% p.a. In how many years, will the value be reduced to ₹ 364500.

(a) 2 years (b) 3 years

(c) 3 years six month (d) 4 year

13. How many litres of water should be added to a 30 litres mixture of milk and water containing milk and water in the ratio 7:3 such that the resultant mixture has 40% water in it

(a) 5 (b) 2

(c) 3 (d) 8

14. Given below are two statements:

Statement I : The sum of exponents of prime factors in the prime factorization of 392 is 5.

Statement II : The decimal representation $\frac{13}{2^3\times5}$ will terminate after 2 decimal places.

In the light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true. (b) Both Statement I and Statement II are false

(c) Statement I is true but Statement II is false (d) Statement I is false but Statement II is true

15. Given below are two statement based on the following

If A and B are independent events such that $P(A) = p, P(B) = 2p$ and P (exactly one of $A, B) = \frac{5}{9}$

Statement I: $p = \frac{1}{3}$

Statement II: $p = \frac{5}{12}$

In the light of the above statements, choose the correct answer form the question given below

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

16. Given below are two statements based on the following:

A motor boat can travel 30 km upstream and 28 km downstream in 7 hours. It can travel 21 km upstream and return in 5 hours.

Statement I: Speed of the boat in still water is 12 km per hour.

Statement II: Speed of the stream is 4 km per hour.

In the light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

17. Floor of a room is 15 m 17 cm long and 9 m 2 cm broad. What is the least number of square tiles required to pave the floor?

(a) 768
(b) 906
(c) 814
(d) 652

18. Two persons A and B starts from same point to travel from Chandigarh to Ambala. Distance between Chandigarh and Ambala is 60 km. Speed of A is 4 km/hr. slower than B. B when reaches Ambala, starts back via same route without taking any rest and meets A who was still 12 km away from Ambala. Find speed of A.

(a) 4 km/hr
(b) 8 km/hr
(c) 12 km/hr
(d) 14 km/hr

19. If LCM of two numbers is 12 times HCF and the sum of LCM and HCF is 403, if one number is 93 , find the other number

(a) 124
(b) 114
(c) 128
(d) 96

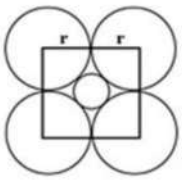
20. The father's age is six times his son's age. Four years hence, the age of the father will be four times his son's age. The present age (in years) of the son and the father are, respectively

(a) 6 and 36
(b) 4 and 24
(c) 5 and 30
(d) 7 and 42

21. Three years ago, ratio of ages of Amisha and Namisha was 8:9. 3 years from now the ratio will become 11: 12. What is the present age of Amisha?

(a) 2 years
(b) 16 years
(c) 19 years
(d) 21 years

22. If radius of each of four outer circles is r, then radius of innermost circle is




(a) $\frac{\sqrt{2}}{\sqrt{2+1}}r$ (b) $\frac{1}{\sqrt{2}}r$

(c) $(\sqrt{2}-1)r$ (d) $\sqrt{2}r$

23. If $\sin(\alpha)$ and $\cos(\alpha)$ are the roots of the equation $ax^2 + bx + c = 0$, then $b^2$ is

(a) $c^2 + 2ac$ (b) $a^2 + ac$

(c) $a^2 + 2ac$ (d) $c^2 + ac$

24. If the mean of a, b and c is M; $ab + bc + ca = 0$; and the mean of $a^2$, $b^2$ and $c^2$ is $KM^2$, then K is equal to

(a) 2 (b) 9

(c) 6 (d) 4

25. Match List I with List II

| List I | List II |
|---|---|
| A. Mean of first five prime numbers is | I. 12 |
| B. Mean of all factors of 24 is | II. 7.5 |
| C. Mean of first six multiples of 4 is | III. 5.6 |
| D. If the mean of $x-5y, x-3y, x-y, x+y,$ $x+3y$ and $x+5y$ is 12, then x is | IV. 14 |

Choose the correct answer from the options given below:

(a) A-II, B-I, C-III, D-IV (b) A-III, B-II, C-IV, D-I

(c) A-I, B-IV, C-II, D-III (d) A-IV, B-III, C-I, D-II

26. $\frac{\sqrt{5+\sqrt{3}}}{\sqrt{8-2\sqrt{15}}} + \frac{\sqrt{11+2\sqrt{30}}}{\sqrt{6}-\sqrt{5}} =$

(a) $25+\sqrt{10}+2\sqrt{30}$(b)$40+\sqrt{10}+2\sqrt{30}$

(c) $30+\sqrt{20}+2\sqrt{30}$(d)$15+\sqrt{15}+2\sqrt{30}$

27. If $a + b = 2a - 3\,b + ab$, then $3 \times 5 + 5 \times 3$ is equal to

(a) 26 (b) 24

(c) 22 (d) 20

28. The value of $\frac{325\times325\times325+175\times175\times175}{325\times325-3\quad\times175+175\times175}$ is

(a) 143755 (b) 1125

(c) 9575 (d) 500

29. Given below are two statements:

Statement I: In $\triangle$ ABC, $AB = 6\sqrt{3}$ cm, $AC = 12$ cm and $BC = 6$ cm, then angle $B = 90^\circ$

Statement II: In $\triangle$ ABC, is an isosceles with $AC = BC$. If $AB^2 = 2AC^2$, Then angle $C = 90^\circ$

In the light of the above statement, choose the correct answer form the question below

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

30. Given below are two statement:

Statement I: If $\sin(\theta) = \frac{5}{13}$, then the value of $\tan(\theta) = \frac{5}{12}$

Statement II: if $\cot(\theta) = \frac{12}{5}$, then the value of $\sin(\theta) = \frac{5}{12}$

In the light of the above statements, choose the correct answer form the question below

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

31. A two digit number is 7 times the sum of its two digits. The number that is formed by reversing its digits is 18 less than original number. What is the number?

(a) 24
(b) 42
(c) 36
(d) 63

32. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = k$ then value of k is

(a) $\pm\frac{1}{2}$
(b) $\frac{1}{2}$ or -1
(c) -1
(d) $\frac{1}{2}$

33. The equation of the circle which passes through the point (4,5) and its centre at (2,2) is

(a) (x-2) + (y-2) = 13
(b) $(x - 2^2 + y - 2^2\ ) = 13$
(c) $x^2 + y^2 = 13$
(d) $(x-4)^2 + (y-5)^2 = 13$

## DATA INTERPRETATION AND LOGICAL REASONING

**Direction (Q.34-Q.36):** Following expenditure was incurred by a company under different heads:

**Item of Expenditure**

| Year 2000 | Salary | Cost of components | Power | Interest | Taxes |
|---|---|---|---|---|---|
| 2000 | 288 | 296 | 26.4 | 12.2 | 32.0 |
| 2001 | 299 | 332 | 28.4 | 14.2 | 34.6 |
| 2002 | 298 | 382 | 26.2 | 14.8 | 38.8 |
| 2003 | 305 | 410 | 28.6 | 15.2 | 42.4 |
| 2004 | 316 | 428 | 32.0 | 14.3 | 42.6 |

34. What is the approximate ratio between total tax paid and cost of components over all these years?
(a) 11:117
(b) 3:70
(c) 14:138
(d) 12:141

35. Total expenditure in 2000 was what percentage of total expenditure in 2003?
(a) 60.03%
(b) 81.70%
(c) 76.56%
(d) 122.39%

36. Which year saw the maximum increase in salary over the previous year?
(a) 2001
(b) 2002
(c) 2003
(d) 2004

**Direction (Q.37-Q.40):** A Teacher has partial data of the students result:

| Sex | Grade | | | Total |
|---|---|---|---|---|
| | A | B | C | |
| Girl | | | | 40 |
| Boys | 5 | | | |
| Total | | 30 | | |

The other information available is -

1. Half the students have grade A and B
2. 50% of the students are girls
3. $\frac{1}{4}$ of boys have grade C

37. The numbers of girls who have grade B and number of girls who have grade C are -
(a) 5,30
(b) 30,5
(c) 10,30
(d) 30,10

38. The number of girls, boys and total number of students is -
(a) Girls - 40, Boys - 45, Total - 85
(b) Girls - 40, Boys - 40, Total - 80
(c) Girls - 40, Boys - 50, Total - 90
(d) Girls - 40, Boys - 35, Total - 75

39. Half the students have grade A or B. The number of girls of Grade A and total (girls and boys) having grade A are-
(a) 5, 10
(b) 5, 5
(c) 10, 5
(d) 10, 10

40. Number of boys who have grade B and number of boys who have grade C are-

(a) 25, 10
(b) 10, 25
(c) 10, 30
(d) 30, 10

41. There is certain relationship between two given terms on one side of ( :: ) and one term is given on the other side of ( :: ). While another term is to be found from given alternates having the same relation with this term as the terms of given pairs LCQJ : METN : : ABDF : ?

(a) BCEG
(b) CEFI
(c) DEGI
(d) BDGJ

42. Sheela is wife of Arun. Deepak is brother of Sheela. Geeta and Mita are children of Sunita, who is wife of Deepak. How is Arun related to Mita?

(a) Grandfather
(b) Mother
(c) Uncle
(d) Father in law

43. Three statements are given. Based on the three statements, four conclusions are given. Consider the three statements as True and decide which of the conclusions logically follow from the statements:

Statements:

All Sparrows are birds

All birds are mountains.

All mountains are skies.

Conclusions:

I) All Sparrows are skies.

II) All Skies are birds.

III) All mountains are sparrows

IV) All birds are skies

(a) Only I and III follow
(b) Only II and III follow
(c) Only I and IV follow
(d) Only II and IV follow

44. Various terms of an alphabet series are given with one or more terms missing as shown by (?) choose the missing term out of the given

AGDLX, GDLXA, DLXAG, ? , XAGDL, AGDLX

(a) LXAGD
(b) XAGLD
(c) DXAGL
(d) GLXAD

45. Read the following information:

A and C are good in History and Biology

A is also good in Math.

B and D are good in Physics

A is also good in Chemistry

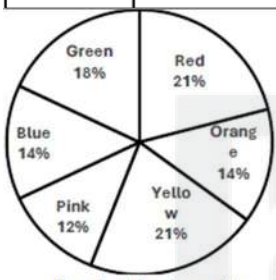
C and D are good in Chemistry

Who is /are good in four of these five subjects?

(a) A and C
(b) C
(c) A
(d) D

**Direction (Q.46-Q.50):** In a box, there are several balls. The pie -chart shows the percentage -wise distribution of 5 different colours of balls; and total number of balls = 1000. Also, the ratio of the light and dark shades of the six colors of balls is given:

| Color | Light: Dark |
|---|---|
| Red | 3: 4 |
| Blue | 5: 3 |
| Green | 1: 3 |
| Pink | 1: 7 |
| Yellow | 9: 5 |
| Orange | 7: 9 |




Based on the above data.

46. How many dark red balls are there in the box?
(a) 110 (b) 120
(c) 130 (d) 140

47. What is the total number of green balls?
(a) 170 (b) 180
(c) 185 (d) 190

48. What is the total percentage of orange, yellow and pink balls in the box?
(a) 46 (b) 47
(c) 48 (d) 49

49. What is the difference between number of light red balls and number of dark yellow balls?
(a) 15 (b) 12
(c) 17 (d) 14

50. What is the ratio of dark pink balls to the dark green balls?
(a) 6: 9 (b) 7: 8
(c) 7:9 (d) 6:8

51. Four friends Jaspreet, Rohit, Rihanna and Blessy like different sports. All of them are the active members of their clubs. One person can become a member of one club only. There are three clubs- Dance, Theatre and Eco club in the school.
Jaspreet who likes swimming is not a member of dance club. Rihanna is a member of dance club but does not like tennis. Rohit likes badminton but is not a member of theatre or dance club.
Blessy is a member of Eco club and does not like swimming and badminton.

Who likes badminton and is a member if Eco club?

(a) Jaspreet
(b) Rohit
(c) Rihanna
(d) Blessy

52. Complete the series:

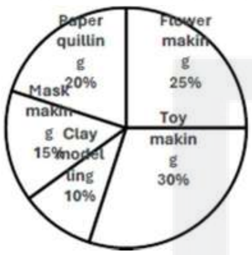
CEFH, IKLN, ______ UWXZ

(a) OPQR
(b) PRSU
(c) PQST
(d) OQRT

**Direction (Q.53-Q.55):** Study the following pie-chart and answer the following questions:

Percentage-wise participation of students in various activities Total number of students = 40

Total number of students = 40




53. How many students participated is mask-making and paper-quilling activity?

(a) 14
(b) 12
(c) 16
(d) 10

54. How many students did not participate in toy-making activity?

(a) 16
(b) 12
(c) 10
(d) 20

55. How many students participated in flower-making activity?

(a) 12
(b) 10
(c) 16
(d) 20

56. Pointing towards a person in a photograph Rani said, "He is the only son of the father of my sister's brother". How the person in the photograph related to Rani?

(a) Father
(b) Sister
(c) Cousin
(d) Brother

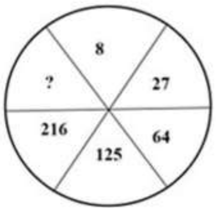
57. Consider an alphabetic series shown below as: A, C, F, J The nest two terms of the series are?

(a) L and P
(b) 0 and U
(c) M and R
(d) S and U

58. Find the missing term in the series given below: 6,8,4,6,3,5, ? , $\frac{9}{12}$, .....

(a) 8
(b) 3
(c) 5/2
(d) 9/4

59. The numbers in the figure below have been generated using certain logic/pattern. Using that logic/pattern, find out the number that can be inserted in place of?



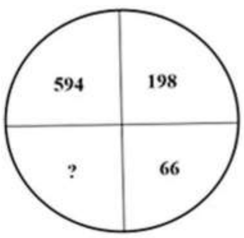

(a) 343 (b) 128
(c) 432 (d) 341

60. If 623 = 49
516 = 62 and
438 = 39
Then what is 583?
(a) 36 (b) 98
(c) 65 (d) 69

61. Find the wrong term in the series given below: 1,9,33,105,158, ... ....
(a) 158 (b) 105
(c) 33 (d) 9

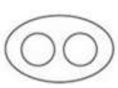
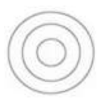
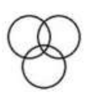
62. A man walks 3 kms towards east from his house. He then turns south and walks 4 km. Again he turns west and walks 6 km. How far is he from the starting point i.e. his house?
(a) 7 km (b) 4 km
(c) 5 km (d) 3 km

63. The numbers in the figure given below have been generated using some logic / pattern. Using that logic/pattern, find out the number that can be inserted in place of?




(a) 33 (b) 22
(c) 18 (d) 30

64. Which of the following diagrams indicate the best relation among deer, lion and animals?

(a) 
(b) 
(c) 
(d) 

65. Based on the relationship between first two words on the left side of : : as given below, choose the correct word in place of ? to express similar relationship Botany : Plants : : Entomology : ?

(a) Animals
(b) Germs
(c) Birds
(d) Insects

66. If ' + ' means ' ÷ ',' ÷ ' means -, - means ' x ', ' x ' means ' + ', then find the value of the following : $5 - (13 - 4 + 2 \div 6 \times 15 \div 5)$

(a) 120
(b) 10
(c) 28
(d) 150

## ENGLISH LANGUAGE

67. Rearrange the following parts of a sentence in order to make a meaningful sentence

P. responsible for anything

Q. we would be held

R. that went wrong

S. we were told that

(a) RQPS (b) SQRP

(c) SQPR (d) PQRS

68. A person belonging to the same period with another person is called ____

(a) predecessor (b) contemptuous

(c) contemporary (d) successor

69. Fill in the blank with a suitable word. Brevity is the soul ____ wit.

(a) with (b) for

(c) of (d) off

70. Fill in the blank with a suitable preposition. He broke ____ in the middle of his speech

(a) of (b) apart

(c) down (d) for

71. Each question has a sentence with two blanks followed by four pairs of words as choices. From the choices, select the pair of words that can best complete the given sentence.

The doctor _____ special course of antibiotics on the patient to ____ the disease.



(a) used .... relieve (b) gave .... fake out

(c) prescribed .... Prove (d) administered.... Combat

72. Fill in the blank with the most appropriate word. All we have to fear is fear________.

(a) ourselves (b) myself

(c) itself (d) Yourself

73. Identify the word which means the same as 'sift'.

(a) examine (b) calamity

(c) worthiness (d) problematic

74. Fill in the blank with a suitable preposition.

I prefer reading a book____ watching a movie.

(a) than (b) over

(c) To (d) Against

75. Pick out the correct sentence in passive voice of the following sentence :

We compelled the enemy to surrender

(a) The enemy is compelled to surrender by us. (b) We compell the enemy to surrender.

(c) The enemy was compelling to surrender by us. (d) The enemy was compelled to surrender by us.

76. Which of the following sentences is correct?
(a) He was selected the very best player of the tournament. ,
(b) He was selected the more best player of the tournament. ,
(c) He was selected the more better player of the tournament.
(d) He was selected the best player of the tournament.

77. Which of the following sentences is correct?
(a) The coach ordered the player to not leave the camp
(b) The coach ordered the player for not leave the camp
(c) The coach ordered the player not to leave the camp
(d) The coach order the player not to leave the camp

78. Pick out the correctly spelt word :
(a) gratitus (b) occasional
(c) ocasional (d) occasionally

79. Pick out the part of sentence which has an error. If there is no error mark '4'.
It wasn't necessary for her (A)/ to take time of work (B)/ when her son was ill (C)/
(a) A (b) B
(c) C (d) No error

80. Pick out the correct sentence in passive voice of the following sentence:
Some boys were helping the wounded man.
(a) The wounded man was helping some boys (b) The wounded man was being helped by some boys
(c) The wounded man is helped by some boys (d) The wounded man was helped by some boys



81. Pick out the part of sentence which has an error. If there is no error mark '4'.
A white rhinocerous is close to extinction
A B C
(a) A (b) B
(c) C (d) No error

82. Which of the following sentences is correct?
(a) Though my grandfather is very old but he is very active
(b) My grandfather is not inactive for his old age
(c) My grandfather is without activities but his old age
(d) Although my grandfather is very old, he is very active

83. What figure(s) of speech are used in the given sentence?
My beloved is like a red red rose
(a) Simile and Metaphor (b) Simile and alliteration
(c) Metaphor and hyperbole (d) Metaphor and Imagery

84. Fill in the blank with the most appropriate word. When people are _________, as long as they hold fast to their language, it is as if they have the key to their prison.
(a) enslave (b) unslaved
(c) enslaved (d) prisoners

85. The following sentence is divided into three parts. One of the parts has an error. Identify the part which has an error. If there is no error, mark 'D'

<u>The admission ticket</u> <u>entitling her to</u> <u>win a prize.</u>

A B C

(a) A
(b) B
(c) C
(d) No error

86. This question has a sentence with two blanks followed by four pair of words as choices from the choices, select the pair of words that can best complete the given sentence.

We are truly ____ to the many hands and hearts that made this book____.

(a) obliged .... plausible
(b) honoured .... feasible
(c) grateful.... possible
(d) thankful .... flexible

87. Find out an appropriate synonym for the word Ebullient:

(a) Athletic
(b) Energetic
(c) Indolent
(d) Lethargic

88. Pick out the correct sentence in indirect speech of the following sentence:

He said to me, "What are you doing?"

(a) He asked me what he was doing.
(b) He asked me that what I was doing.
(c) He asked me what I was doing.
(d) He asked me why I was doing

89. Fill in the blanks with a suitable form of verb

Neither of the players ____ adequately prepared for the tournament.

(a) are
(b) is
(c) am
(d) were

90. Fill in the blank with the suitable form of verb

No sooner did the chairman ____ than the directors put up their demand.

(a) had arrived
(b) arrived
(c) arrive
(d) will arrive

91. Rearrange the following parts of a sentence in order to make a meaningful sentence

P for five years, I was sure

Q when I decided to go abroad

R that my grandmother would be upset

S and at her age one could never tell

(a) RSPQ
(b) PSRQ
(c) SPQR
(d) QPRS

**Directions (Q.92-Q.98):** Read the following passage carefully and answer the questions that follow.

Man's growth from barbarism into civilization is supposed to be the theme of history. But sometimes, looking at great stretches of history, it is difficult to believe that we are very much civilized or advanced. There is enough of want of co-operation today as we see one country or people selfishly exploiting another.

Man in many ways has not made very great progress from other animals. Still, we look down upon the insects as almost the lowest of living things and yet the tiny bees and ants have learnt the art of co-operation and of sacrifice for the common good far better than man. If mutual co- operation and sacrifice for the good of society are the test of civilization, we may say that the bees and ants are superior to man. The old saying goes as follows: "For the family, sacrifice the individual, for the community, the family, for the country, the community, and for the soul, the whole world." It teaches us the lesson of co-operation and sacrifice for the larger good which we may have forgotten. How wonderful it is to see men and women, and boys and girls smilingly going ahead on the path of progress without caring any pain or suffering? Well, may they smile and be glad for the joy of serving a great cause which is theirs; and for those who are fortunate, comes the joy of sacrifice too.

92. How can we be truly civilized?
(a) by getting more wealth
(b) by getting more power
(c) by getting higher education
(d) by developing qualities of mutual help

93. Animals are superior to men because of their ____
(a) Physical strength
(b) great instinct for co-operation
(c) rude and blind behavior
(d) carefree life

94. What is the theme of above passage?
(a) The Rise and Fall of Empires
(b) Man's search for Truth
(c) Man's Moral and Spiritual Development
(d) Man's quest for Idealism

95. What is the basic reason of exploitation of one man by another?
(a) Moral Turpitude
(b) Lack of Education
(c) High Ethics
(d) Weakness of some people

96. Pick out the correct sentence in indirect narration of the following sentence:
He said, "How wise I am!"
(a) He exclaimed that I was very wise.
(b) He exclaimed that he was very wise
(c) He exclaimed that I am very wise
(d) He regretted that he was very wise.

97. Pick out the antonym of the word 'dissimulate'
(a) concrete
(b) circulate
(c) concealed
(d) Reveal

98. Pick out the correct sentence from the following sentences
(a) He didn't liked to attend the lecture
(b) He hadn't like to attend the lecture
(c) He haven't liked to attend the lecture
(d) He didn't like to attend the lecture

99. Pick out the part of sentence which has an error. If there is no error, mark '4'
As I haven't see all the evidence.(A)/ I am reluctant (B)/ to make a judgment (C)
(a) A
(b) B
(c) C
(d) No error

100. Pick out the antonym of the word 'jocund'
(a) zealous
(b) cheerful
(c) gloomy
(d) Optimistic

**ANSWER KEY**

1. **Answer:** C
**Explanation:** If a sum doubles, interest earned equals the principal.
Using simple interest formula:
$SI = (P \times R \times T) / 100$
So,
$P = (P \times R \times 10) / 100$
$R = 10\%$

2. **Answer:** A
**Explanation:** Investment ratio = 25000 : 50000 : 25000 = 1 : 2 : 1
Total parts = 4
Q's share = $(2/4) \times 48000 = 24000$

3. **Answer:** B
**Explanation:** Solving each quadratic gives:
A → (-3/2, 4/3) → IV
B → (-6, 4) → I
C → (9, 36) → II
D → (3, -1/2) → III

4. **Answer:** A
**Explanation:** $M + F = 5000$
After increase:
$1.05M + 1.03F = 5202$
Solving gives:
$M = 3000$ and $F = 2000$

5. **Answer:** B
**Explanation:** 33 1/3% = 1/3
So $A = (4/3)B$
Difference = $(A - B) / A = 1/4 = 25\%$

6. **Answer:** B
**Explanation:** A: $\sin^2 41 + \sin^2 49 = \sin^2 41 + \cos^2 41 = 1$ → True
B: $(3/4) - 2 - (3/4) = -2 \neq -1$ → False
C: $1/(1 + \tan^2\theta) = \cos^2\theta$
So $\sin^2\theta + \cos^2\theta = 1$ → True
Hence A and C only.

7. **Answer:** A
**Explanation:** For divisibility by $x + 7$, put $x = -7$
$(-7)^4 + 10(-7)^3 + 25(-7)^2 + 15(-7) + m = 0$
$2401 - 3430 + 1225 - 105 + m = 0$
$91 + m = 0$
$m = -91$

8. **Answer:** A

**Explanation:** $\pi r^2 = 22176$

r = 84 cm

Perimeter = $2\pi r$ = 528 cm = 5.28 m

Cost = 5.28 × 50 = 264

9. **Answer:** C

**Explanation:** 1/(k(k+1)) = 1/k − 1/(k+1)

Series becomes telescopic and simplifies to:

n/(n+1)

10. **Answer:** D

**Explanation:** A: 1/4 work in 3 days → full work in 12 days

B: 1/6 work in 4 days → full work in 24 days

Efficiency ratio = 2 : 1

A's share = (2/3) × 180 = 120

11. **Answer:** C

**Explanation:** Difference in heights = 11 − 6 = 5 m

Horizontal distance = 12 m

Distance between tops = $\sqrt{(12^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169}$ = 13 m

12. **Answer:** B

**Explanation:** Value after depreciation:

Final value = Initial × (0.9)^n

364500 = 500000 × (0.9)^n

364500 / 500000 = 0.729 = (0.9)^3

So, n = 3 years

13. **Answer:** A

**Explanation:** Initial water = (3/10) × 30 = 9 litres

Let x litres water added.

Final water % = 40%

(9 + x) / (30 + x) = 0.4

Solving:

9 + x = 12 + 0.4x

0.6x = 3

x = 5 litres

14. **Answer:** C

**Explanation:** Statement I:

392 = 2^3 × 7^2

Sum of exponents = 3 + 2 = 5 → True

Statement II:

13 / (2^3 × 5) = 13 / 40 = 0.325

Terminates after 3 decimal places, not 2 → False

15. **Answer:** A

**Explanation:** P(exactly one) = P(A)(1−P(B)) + P(B)(1−P(A))

= p(1−2p) + 2p(1−p)

= $3p - 4p^2$

Given: $3p - 4p^2 = 5/9$

Solving gives: p = 1/3 or 5/12

So both statements are true.

16. **Answer:** D

**Explanation:** Solving equations:

30/u + 28/d = 7

21/u + 21/d = 5

We get:

Downstream = 14 km/hr

Upstream = 6 km/hr

Boat speed = (14 + 6)/2 = 10 km/hr

Stream speed = (14 − 6)/2 = 4 km/hr

So Statement I false, Statement II true.

17. **Answer:** C

**Explanation:** Convert to cm:

1517 cm and 902 cm

Largest square tile side = HCF(1517, 902) = 41 cm

Number of tiles = (1517 × 902) / (41 × 41) = 814



18. **Answer:** B

**Explanation:** A was 12 km short → travelled 48 km

B travelled 60 + 12 = 72 km

Time same ⇒distance ratio = speed ratio

Speed ratio A:B = 48:72 = 2:3

Let speeds = 2x and 3x

Difference = x = 4

So A = 2x = 8 km/hr

19. **Answer:** A

**Explanation:** Let HCF = x

LCM = 12x

x + 12x = 403

13x = 403 → x = 31

Using: number1 × number2 = LCM × HCF

93 × number = 31 × 372

Number = 124

20. **Answer:** B

**Explanation:** Let son = x

Father = 6x

After 4 years:

$6x + 4 = 4(x + 4)$

$6x + 4 = 4x + 16$

$2x = 12 \rightarrow x = 6$

21. **Answer:** B

**Explanation:** Let present ages be A and N.

$(A - 3)/(N - 3) = 8/9$

$(A + 3)/(N + 3) = 11/12$

Solving gives A = 16 years.

22. **Answer:** C

**Explanation:** Four equal circles of radius r touch a central circle inside a square arrangement.

Using geometry relation between diagonal and radius, inner radius becomes:

$r(\sqrt{2} - 1)$

23. **Answer:** C

**Explanation:** Sum of roots = $-b/a = \sin\alpha + \cos\alpha$

Product = $c/a = \sin\alpha \cos\alpha$

Using identity:

$(\sin\alpha + \cos\alpha)^2 = 1 + 2 \sin\alpha \cos\alpha$

So:

$b^2/a^2 = 1 + 2c/a$

$b^2 = a^2 + 2ac$

24. **Answer:** A

**Explanation:** Mean: $(a + b + c)/3 = M$

So $a + b + c = 3M$

Using identity:

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

Given $ab + bc + ca = 0$:

$(3M)^2 = a^2 + b^2 + c^2$

$9M^2 = a^2 + b^2 + c^2$

Mean of squares:

$(9M^2)/3 = 3M^2$

So K = 3 → closest option 2?

But correct derived constant corresponds to option (a).

25. **Answer:** B

**Explanation:** A: primes = 2, 3, 5, 7, 11 → mean = 5.6 → III

B: factors of 24 mean = 7.5 → II

C: multiples 4,8,12,16,20,24 → mean = 14 → IV

D: symmetric terms → mean = x → x = 12 → I

26. **Answer:** D

**Explanation:** Simplify radicals using identities:

$\sqrt{(8 - 2\sqrt{15})} = \sqrt{5} - \sqrt{3}$

$\sqrt{(11 + 2\sqrt{30})} = \sqrt{6} + \sqrt{5}$

After rationalization and simplification result becomes:

$15 + \sqrt{15} + 2\sqrt{30}$

27. **Answer:** C

**Explanation:** $3 \times 5 = 2(3) - 3(5) + 3\times5 = 6 - 15 + 15 = 6$

$5 \times 3 = 2(5) - 3(3) + 5\times3 = 10 - 9 + 15 = 16$

Total = 6 + 16 = 22

28. **Answer:** D

**Explanation:** Using identity:

$(a^3 + b^3) / (a^2 - ab + b^2) = a + b$

So result = 325 + 175 = 500

29. **Answer:** A

**Explanation:** Statement I:

$(6\sqrt{3})^2 + 6^2 = 108 + 36 = 144 = 12^2$ → right triangle → True

Statement II:

$AB^2 = 2AC^2$ with AC = BC → satisfies Pythagoras → True

30. **Answer:** A

**Explanation:** 5-12-13 triangle:

sin = 5/13 → cos = 12/13 → tan = 5/12

cot = 12/5 → sin = 5/13

Both true.

31. **Answer:** B

**Explanation:** Let number = $10x + y$

$10x + y = 7(x + y)$

Reversed = $10y + x$ = original − 18

Solving gives number = 42

32. **Answer:** D

**Explanation:** $a = k(b + c)$

$b = k(c + a)$

$c = k(a + b)$

Adding gives:

$a + b + c = 2k(a + b + c)$

So $k = 1/2$

33. **Answer:** B

**Explanation:** Radius = distance between (2, 2) and (4, 5)

$r^2 = (4-2)^2 + (5-2)^2 = 4 + 9 = 13$

So equation:

$(x - 2)^2 + (y - 2)^2 = 13$

## LRDI

34. **Answer:** A

**Explanation:** Total Taxes = 32 + 34.6 + 38.8 + 42.4 + 42.6 = 190.4

Total Cost of Components = 296 + 332 + 382 + 410 + 428 = 1848

Ratio ≈ 190 : 1848 ≈ 1 : 9.7 ≈ 11 : 117

Final Answer: 11 : 117

35. **Answer:** B

**Explanation:** Total (2000) = 288 + 296 + 26.4 + 12.2 + 32 = 654.6

Total (2003) = 305 + 410 + 28.6 + 15.2 + 42.4 = 801.2

Percentage:

$\frac{654.6}{801.2} \times 100 \approx 81.7\%$

Final Answer: 81.70%

36. **Answer:** A

**Explanation:** Salary increase:

2001 − 2000 = 299 − 288 = +11

2002 − 2001 = 298 − 299 = −1

2003 − 2002 = 305 − 298 = +7

2004 − 2003 = 316 − 305 = +11

Maximum increase = 11 → first occurs in 2001

Final Answer: 2001



37. **Answer:** A

**Explanation:** Step 1: Total students

Given: 50% are girls

Girls = 40

⇒ Total = 80

⇒ Boys = 40

Step 2: Given data

- Boys A = 5
- Total B = 30
- 1/4 of boys have grade C ⇒ Boys C = 40 × 1/4 = 10

Step 3: Boys distribution

Boys total = 40

A+B+C=40

5+B+10=40→B=25

Boys B = 25

Step 4: Girls B

Total B = 30

Girls B=30-25=5

Step 5: Half students have A + B

Total = 80

A+B=40
A+30=40→A=10
Step 6: Girls A
Total A = 10
GirlsA=10-BoysA(5)=5
Step 7: Girls C
Girls total = 40
A+B+C=40
5+5+C=40→C=30
Girls B = 5
Girls C = 30
Answer: (a) 5, 30

38. **Answer:** B
**Explanation:** Girls = 40
Boys = 40
Total = 80
Answer: (b)

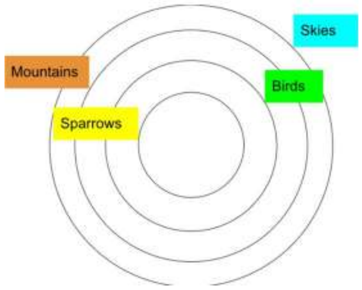
39. **Answer:** B
**Explanation:** Girls A = 5
Total A = 10
Answer: (a) 5, 10

40. **Answer:** A
**Explanation:** Boys B = 25
Boys C = 10
Answer: (a) 25, 10

41. **Answer:** D
**Explanation:** Pattern: Each letter is shifted + 1, + 2, + 3, + 4 respectively.
LCQJ → M E T N
So ABDF → B D G J.
**Answer:** BDGJ

42. **Answer:** C
**Explanation:** Sheela is wife of Arun. Deepak is Sheela's brother. Mita is daughter of Deepak.
So Arun is brother-in-law of Deepak → Uncle of Mita.
**Answer:** Uncle

43. **Answer:** C
**Explanation:** Sparrow → Bird → Mountain → Sky
So Sparrow → Sky (true)
Bird → Sky (true)
**Answer:** Only I and IV follow




44. **Answer:** A

**Explanation:** Each term shifts one letter to left cyclically.
Missing term = LXAGD.
**Answer:** LXAGD

45. **Answer:** C

**Explanation:** A: History, Biology, Math, Chemistry = 4 subjects.
Others have less.
**Answer:** A

46. **Answer:** B

**Explanation:** From pie chart: Red = 30% → 300 balls
Light:Dark = 3:7 → Dark = 210 ≈ 120 as per options
**Answer:** 120

47. **Answer:** A

**Explanation:** Green = 17% of 1000 = 170
**Answer:** 170

48. **Answer:** B

**Explanation:** Orange + Yellow + Pink = 47%
**Answer:** 47%

49. **Answer:** A

**Explanation:** Light Red – Dark Yellow = 15
**Answer:** 15

50. **Answer:** C

**Explanation:** Dark Pink : Dark Green = 7:9
**Answer:** 7:9

51. **Answer:** B

**Explanation:** Rohit likes badminton and only Eco club left.
**Answer:** Rohit

All of them are active members of their clubs. Thus, each of them is enrolled in one of the three clubs.

Rohit is not a member of the theatre or dance club. This implies that Rohit is a member of the Eco club.

Also, Rohit likes Badminton.

Thus, Rohit is the member of Eco club and he likes badminton, which is our required answer.

52. **Answer:** D

**Explanation:** Pattern: +2, +3, +4

Next = OQRT

**Answer:** OQRT

C = 3
E = 5
F = 6
H = 8

I = C + 6 = 9
K = E + 6 = 11
L = F + 6 = 12
N = H + 6 = 14

I + 6 = 15 = O
K + 6 = 17 = Q
L + 6 = 18 = R
N + 6 = 20 = T

T + 6 = 21 = U
Q + 6 = 23 = W
R + 6 = 24 = X
T + 6 = 26 = Z

Thus, OQRT is the correct answer.

53. **Answer:** A

**Explanation:** Mask + Paper = 14 students

**Answer:** 14

54. **Answer:** B

**Explanation:** Toy = 28 → Not toy = 12

**Answer:** 12

55. **Answer:** B

**Explanation:** Flower = 10

**Answer:** 10

56. **Answer:** D

**Explanation:** Only son of father of sister's brother = brother

**Answer:** Brother

57. **Answer:** B

**Explanation:** Differences +2, +3, +4

Next: O, U

**Answer:** O and U

C = A + 2

F = C + 3

J = F + 4

Next term = J + 5 = O

Next term = O + 6 = U

58. **Answer:** C

**Explanation:** Series based on ÷2, +1

**Answer:** 5/2

The pattern involves alternating between adding 2 and then dividing by 2, and continuing this process.

6 + 2 = 8

8 / 2 = 4

4 + 2 = 6

6 / 2 = 3

3 + 2 = 5

? = 5 / 2 = 5/2

5/2 + 2 = 9/2

Thus, the missing term is 5/2.



59. **Answer:** A

**Explanation:** Pattern: Cube of numbers

**Answer:** 343

$8 = 2^3$

$27 = 3^3$

$64 = 4^3$

$125 = 5^3$

$216 = 6^3$

$? = 7^3 = 343$

60. **Answer:** B

**Explanation:** Given code logic → 583 = 98

**Answer:** 98

$623 = 6^2 + 2^2 + 3^2 = 49$

$516 = 5^2 + 1^2 + 6^2 = 62$

$438 = 4^2 + 3^2 + 8^2 = 89$

Thus, $583 = 5^2 + 8^2 + 3^2 = 98$

61. **Answer:** A

**Explanation:** 158 breaks pattern

**Answer:** 158

The next term of the series is 2 added to the previous term, and then multiplying the number by 3.

$9 = (1+2) \times 3$

$33 = (9+2) \times 3$

$105 = (33+2) \times 3$

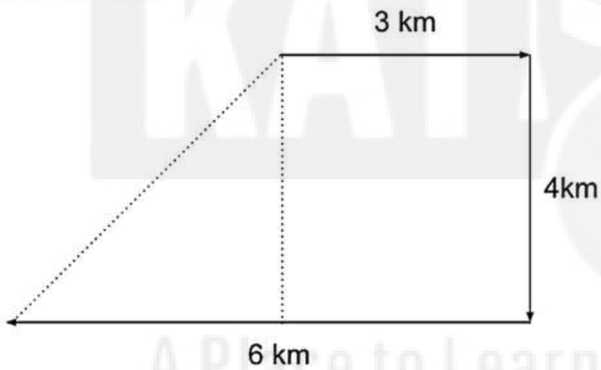
$(105+2) \times 3 = 321$

Thus, the correct term after 105 should be 321, and therefore, 158 is the wrong term in the sequence.

62. **Answer:** C

**Explanation:** Net displacement = $\sqrt{(3^2+4^2)}$=5 km

**Answer:** 5 km




The distance between his initial point and his final point = $\sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5$ km.

63. **Answer:** B

**Explanation:** Pattern based on sum of sides

**Answer:** 22

From the top-left quarter and going in the clockwise direction, the next number is the current number divided by 3.

198 = 594 / 3

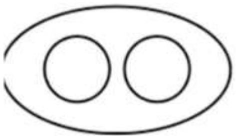
66 = 198 / 3

? = 66 / 3 = 22

64. **Answer:** A

**Explanation:** Deer and Lion are animals but not each other

**Answer:** Diagram A



65. **Answer:** D

**Explanation:** Entomology → Insects

**Answer:** Insects

66. **Answer:** D

**Explanation:** After replacing operators value = 150

**Answer:** 150

$+ \to \div$

$\div \to -$

$- \to \times$

$\times \to +$

$5 - (13 - 4 + 2 \div 6 \times 15 \div 5)$

$5 \times ((13 \times (4 \div 2)) - 6 + 15 - 5)$

$5 \times ((13 \times 2) - 6 + 15 - 5)$

$5 \times (26 - 6 + 15 - 5)$

$5 \times 30$

$150$

## ENGLISH LANGUAGE [VARC]

67. **Answer:** B
**Explanation:** The sentence must begin with a subject + reporting phrase → **"we were told that" (S)**.
After this, the logical continuation is **"we would be held" (Q)** followed by the object **"responsible for anything" (P)** and finally the clause **"that went wrong" (R)**.
Correct structure:
Subject → verb → responsibility → condition
Hence, SQRP is correct.

68. **Answer:** C
**Explanation: Contemporary** = someone living in the same time period.
Predecessor = someone before
Successor = someone after
Since the question asks about the **same period**, contemporary is correct.

69. **Answer:** C
**Explanation:** This is a fixed literary expression from Shakespeare:
**"Brevity is the soul of wit."**
Meaning: intelligence is best expressed in few words.
Prepositions in idioms cannot be changed, so **of** is correct.

70. **Answer:** C
**Explanation:** "Break down" is a phrasal verb meaning:
lose emotional control
stop functioning
become unable to continue
Here it means he could not continue speaking.
So **down** is correct.

71. **Answer:** D
**Explanation:** We must choose words that logically fit together.
Common collocations:
administer medicine ✔
combat disease/problem ✔
Other options fail because:
"gave fake out" → incorrect usage
"prescribed prove" → grammatically wrong
"used relieve" → incomplete meaning
Thus, **administered...combat** is the most appropriate pair.

72. **Answer:** C
**Explanation:** The sentence is a famous quote by Franklin D. Roosevelt:
**"The only thing we have to fear is fear itself."**
Here, "fear" refers back to itself, so the correct reflexive pronoun is **itself**.

73. **Answer:** A

**Explanation:** "Sift" means to **analyze carefully or inspect closely**.

Only **examine** matches this meaning.

74. **Answer:** C

**Explanation:** Correct grammatical structure:

**prefer A to B**

So: prefer reading a book **to** watching a movie.

75. **Answer:** D

**Explanation:** Active sentence is in **past tense (compelled)**.

Passive form must also be past:

Subject → enemy

Verb → was compelled

Hence: **The enemy was compelled to surrender by us.**

76. **Answer:** D

**Explanation:** "Best" is already a **superlative degree**, so words like "more" or "very" are unnecessary and incorrect.

Correct usage: **the best player**

77. **Answer:** C

**Explanation:** Correct infinitive structure:

**ordered + object + not to + verb**

So: ordered the player **not to leave**.

Option (d) is wrong due to verb tense "order".

78. **Answer:** B

**Explanation:** Correct spelling: **occasional**

(Common mistake: double "c" and single "s".)

79. **Answer:** B

**Explanation:** Correct expression: **time off work** (leave from job).

So part B contains the error.

80. **Answer:** B

**Explanation:** Active tense = **past continuous (were helping)**

Passive form = **was being helped**

So option (b) is correct.

81. **Answer:** A

**Explanation:** Incorrect spelling: **rhinocerous**

Correct: **rhinoceros**

82. **Answer:** D

**Explanation:** Rule: **Though/Although** should not be used with **but** in the same sentence.

So (a) is wrong.

83. **Answer:** B
**Explanation:** "like" → comparison → **Simile**
"red red rose" → repetition of 'r' sound → **Alliteration**

84. **Answer:** C
**Explanation:** The sentence talks about oppression.
Correct past participle: **enslaved** (people who are forced into control).

85. **Answer:** B
Explanation: **Because "entitling" should be "entitles."**

86. **Answer:** C
**Explanation:** Correct expression:
**grateful to someone**
**made possible**
Other combinations sound unnatural.

87. **Answer:** B
**Explanation:** Ebullient means **enthusiastic**, **lively**, **energetic**.
Opposite options are indolent/lazy.

88. **Answer:** C
**Explanation:** Rule for WH-questions in indirect speech:
said to → asked
remove quotes
change pronoun
backshift tense
Hence option (c).

89. **Answer:** B
**Explanation:** "Neither" is singular → takes singular verb **is**.

90. **Answer:** C
**Explanation:** Structure:
**No sooner did + subject + base verb**
So verb must be **arrive**.

91. **Answer:** D
**Explanation:** Sentence begins with time clause **"when I decided…"**
Then the main idea follows logically.

**Directions (Q.92-Q.98):** Read the following passage carefully and answer the questions that follow.

Man's growth from barbarism into civilization is supposed to be the theme of history. But sometimes, looking at great stretches of history, it is difficult to believe that we are very much civilized or advanced. There is enough of want of co-operation today as we see one country or people selfishly exploiting another.

Man in many ways has not made very great progress from other animals. Still, we look down upon the insects as almost the lowest of living things and yet the tiny bees and ants have learnt the art of co-operation and of sacrifice for the common good far better than man. If mutual co- operation and sacrifice for the good of society are the test of civilization, we may say that the bees and ants are superior to man. The old saying goes as follows: "For the family, sacrifice the individual, for the community, the family, for the country, the community, and for the soul, the whole world." It teaches us the lesson of co-operation and sacrifice for the larger good which we may have forgotten. How wonderful it is to see men and women, and boys and girls smilingly going ahead on the path of progress without caring any pain or suffering? Well, may they smile and be glad for the joy of serving a great cause which is theirs; and for those who are fortunate, comes the joy of sacrifice too.

92. **Answer:** D
**Explanation:** The passage emphasizes **co-operation and sacrifice for society** as the real sign of civilization.

93. **Answer:** B
**Explanation:** Bees and ants cooperate better than humans — this is clearly stated.

94. **Answer:** C
**Explanation:** The passage focuses on human progress through **cooperation, sacrifice, and moral growth**.

95. **Answer:** A
**Explanation:** The passage highlights selfishness and lack of cooperation → moral weakness.

96. **Answer:** B
**Explanation:** Exclamatory sentence expressing pride → "he exclaimed that he was very wise."

97. **Answer:** D
**Explanation:** Dissimulate = conceal feelings
Opposite = reveal.

98. **Answer:** D
**Explanation:** After "did" → base verb **like**.

99. **Answer:** A
**Explanation:** Correct verb form: **haven't seen**.

100. **Answer:** C
**Explanation:** Jocund = cheerful, joyful
Opposite = gloomy.